

- A robot can move between 2 rooms R_0 and R_1 as follows:



- i.e.,
- (a) it starts in room R_0
 - (b) when in room R_0 it can only move to room R_1
 - (c) when in room R_1 it can move to R_1 or R_0

- To model this system, let us consider the following atomic propositions: $p_0, p_1, p_2, \dots$
such that p_i means that the robot is in room R_0 at step i (i.e., after i moves)
and $\neg p_i$ means that the robot is in room R_1 at step i
- we can model the initial state of the robot as follows: p_0 , i.e., the robot is initially in room R_0 (a)
- we can model a transition as follows: $(p_i \rightarrow \neg p_{i+1}) \wedge (\neg p_i \rightarrow (\neg p_{i+1} \vee p_{i+1}))$, i.e., (b) $\wedge$ (c)
- we can capture two moves as follows: $\underbrace{p_0}_{\text{initial position}} \wedge \underbrace{(p_0 \rightarrow \neg p_1)}_{\text{1st move}} \wedge \underbrace{(\neg p_0 \rightarrow (\neg p_1 \vee p_1))}_{\text{2nd move}} \wedge \underbrace{(p_1 \rightarrow \neg p_2)}_{\text{3rd move}} \wedge \underbrace{(\neg p_1 \rightarrow (\neg p_2 \vee p_2))}_{\text{4th move}}$
- Could we prove that when in room R_1 , the robot will be in room R_1 next?
 $\neg$ No: when in R_1 , the robot can move to R_0 $\leftarrow$ call it P
- How do we formally prove that this property fails? we prove its negation, i.e. $\neg(\neg p_i \rightarrow \neg p_{i+1})$
- let's show that this fails after 2 steps, i.e.: $p_0 \wedge (p_0 \rightarrow \neg p_1) \wedge (\neg p_0 \rightarrow (\neg p_1 \vee p_1)) \wedge (p_1 \rightarrow \neg p_2) \wedge (\neg p_1 \rightarrow (\neg p_2 \vee p_2)) \wedge \neg(\neg p_2 \rightarrow \neg p_3)$
- We'll show this using a SAT solver: (1) convert the formula to a CNF using logical equivalences
(2) use DPLL to check whether it is satisfiable

$$\begin{aligned}
 (1) & p_0 \wedge (p_0 \rightarrow \neg p_1) \wedge (\neg p_0 \rightarrow (\neg p_1 \vee p_1)) \wedge (p_1 \rightarrow \neg p_2) \wedge (\neg p_1 \rightarrow (\neg p_2 \vee p_2)) \wedge \neg(\neg p_2 \rightarrow \neg p_3) \\
 \Leftrightarrow & p_0 \wedge (\neg p_0 \vee \neg p_1) \wedge (p_0 \vee \neg p_1 \vee p_1) \wedge (\neg p_1 \vee \neg p_2) \wedge (p_1 \vee \neg p_2 \vee p_2) \wedge \neg(\neg p_2 \vee \neg p_3) \quad - \text{elim } \rightarrow \\
 \Leftrightarrow & p_0 \wedge (\neg p_0 \vee \neg p_1) \wedge (p_0 \vee \neg p_1 \vee p_1) \wedge (\neg p_1 \vee \neg p_2) \wedge (p_1 \vee \neg p_2 \vee p_2) \wedge \neg \neg p_2 \wedge p_3 \quad - \text{de Morgan} \\
 \Leftrightarrow & p_0 \wedge (\neg p_0 \vee \neg p_1) \wedge (p_0 \vee \neg p_1 \vee p_1) \wedge (\neg p_1 \vee \neg p_2) \wedge (p_1 \vee \neg p_2 \vee p_2) \wedge p_2 \wedge p_3 \quad - \neg \text{elim}
 \end{aligned}$$

CNF $\rightarrow$

$$(2) \text{ use DPLL: } \cancel{p_0} \wedge \cancel{(\neg p_0 \vee \neg p_1)} \wedge \cancel{(p_0 \vee \neg p_1 \vee p_1)} \wedge \cancel{(\neg p_1 \vee \neg p_2)} \wedge \cancel{(p_1 \vee \neg p_2 \vee p_2)} \wedge \cancel{p_2} \wedge p_3$$

$p_0 = T$ $p_1 = F$ $p_2 = T$: the formula is satisfiable

This gives us a run of our robot: $R_0 \rightarrow R_1 \rightarrow R_1$, which shows that the property P does not hold, i.e., we obtained a counterexample.